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E-232 CONDUITS AND FITTINGS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of conduits and conduit fittings for auxiliary power, control and instrumentation cables in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5, Codes and Standards, and the following specific standards:

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 American National Standards Institute

ANSI C33.65 : Conduits and Boxes

ANSI C80.1 : Rigid Steel Conduits-Zinc Coated

ANSI C80.3 : Specifications for Electrical Metallic Tubing-Zinc Coated

ANSI C80.4 : Fittings for Rigid Metal Conduit and Electrical Metallic Tubing.

2.3 American Society of Test Materials

ASTM A 653 : Standard Specifications for Steel Sheet, Zinc Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.

2.4 National Electrical Manufacturers Association

NEMA ML1 : Metal Framing (Continuous Slot Metal Channel Systems)

2.5 National Fire Protection Association

NFPA 70 : National Electrical Code.

3. OPERATING AND DESIGN CONDITIONS

The conduits shall be suitable for an outdoor installation in a gas fired combined-cycle, power plant in a tropical climate. The Site meteorological, seismic and environmental conditions are described in Part 4, Subsections 1.1, 2.14 and 2.16, respectively.

4. DETAILED REQUIREMENTS

4.1 General

The Contractor shall furnish all conduits, conduit bends, fittings, pull box devices, cabinets, junction boxes, supports, clamps, and all other conduit appurtenances. All such materials shall comply with these Specifications.

Conduits that will be concealed, in walls or embedded in concrete shall be shown on the Contractor's electrical installation drawings.

4.2 Conduit Sizing

The following data shall be used for determining the sizes of the conduits for the various conditions:

- (1) The cross sectional area of a single-cable in a conduit shall not occupy more than 53% of the cross sectional area of the conduit.
- (2) When two (2) or more cables are to be installed in a single-conduit, the total of the cross sectional areas of the cables shall not occupy more of the cross sectional area of the conduit than is shown below:
 - a. 2 cables more than 30 %
 - b. 3 cables more than 40 %
 - c. 4 cables more than 38 %
 - d. over 4 cables more than 35 %

The area of ground cables inside the conduit shall be included in the cable area.

Short straight runs of conduit and short conduit terminations may have a greater percentage of the total cross section area occupied than that indicated above, up to a maximum of 60 %.

4.3 Conduits and Fittings

Unless otherwise indicated, all conduits and fittings for indoor or outdoor use shall be galvanized rigid steel. Conduits for indoor lighting in non-hazardous areas shall be electrical metallic tubing (EMT).

4.3.1 Steel Conduits and Fittings

All steel conduits, couplings, nipples, and elbows (bends) specified, shall conform to ANSI Standard C80.1 Rigid Steel Conduit-Zinc Coated, except that minimum elbow (bend) radii shall be those radii dimensions shown in Clause 4.4.2.

All fittings for steel conduits shall be either steel or malleable iron, except for the insulated bushings which are covered in Clause 4.3.4. Such metal fittings shall be hot dipped galvanized, in accordance with Clause 4.3.6. Cast iron or light alloy die-cast metal fittings shall not be used.

4.3.2 Electrical Metallic Tubing (EMT) and Fittings

All steel EMT and EMT bends shall be electro-galvanized or equivalent, and shall comply with the latest ANSI Standard C80.3.

All steel EMT fittings, or the steel portions of such fittings, shall be electro galvanized, cadmium plated, or otherwise protected with an approved rust preventative; and shall comply with the latest ANSI Standard C80.4 or equivalent.

All couplings and connectors for EMT shall be compression type to ensure positive grounding continuity. Indenter type or die cast fittings shall not be used.

4.3.3 Flexible Conduits

Flexible conduit shall be used to dampen vibration or where relative movement is required between the conduit and connected apparatus, as at motors. All flexible conduits shall be liquidtight, galvanized steel.

Liquidtight flexible metal conduits which serve power equipment as a ground fault path shall be provided with fittings and bonding jumpers.

Liquid tight flexible metal conduits which serve instrument and control circuits shall be provided with liquidtight nylon insulated connectors and sealing rings, except liquidtight flexible conduit 40 mm or smaller shall be provided with cable grips.

4.3.4 Insulated Bushings

Insulated bushings at conduit terminations shall be either wholly or partially of phenolic material, nylon or fiber, and shall be highly resistant to breakage. Bushings made of thermoplastic materials; such as, polyethylene, cellulose acetate butyrate, or other flammable materials are not acceptable and shall not be used.

4.3.5 Conduit Supports and Fasteners

All conduit supports and fasteners, such as, pipe straps, U-Bolts, J-Bolts, etc., shall be hot dipped galvanized, electro-galvanized or cadmium plated.

4.3.6 Conduits, Fittings and Enclosures for Various Locations

All conduits shall be hot dipped galvanized rigid steel conduits except as noted herein.

Where steel conduits and fittings are installed outdoors, in underground duct banks or embedded in concrete floors where the inner conduit surface may be subjected to the presence of water, the conduits and fittings shall be hot dipped galvanized rigid steel, with hot dipped galvanizing applied to both inner and outer surfaces, including the threads.

Hazardous areas shall have the equipment installed in conformance with Article 500 of NFPA 70, National Electrical Code requirements for the particular hazard.

4.4 Installation of the Conduit System

4.4.1 Conduits

All field cut conduits shall be cut square and then reamed to remove any sharp edges.

Conduits shall be threaded according to the relevant standards.

No more than one power cable shall be installed in a conduit unless they are feeding the same service. If more than one control or instrument cable is to be routed to a single piece of equipment, the Contractor may, at his discretion, run the cables (except low-level instrumentation cables and cables that must not be combined with other cables in common conduits or more than one power cable in the same conduit) in a single conduit of proper size. If a group of control or instrument cables are running to the same general area, the Contractor may run all cables through a single conduit to a junction box conveniently located so that single conduits of comparatively short length can be run to their individual destinations.

Conduits shall not be rigidly connected to a motor, cubicle or piece of equipment that is provided with flexible mountings.

The following minimum clearances shall be maintained:

- (1) Conduits shall have an 150 mm minimum clearance from the outside surface of the insulation on any insulated steam pipe in its thermally expanded position.
- (2) Junction boxes, pull boxes, termination cabinets, control stations, etc., shall have a 500 mm minimum clearances from the outside surface of the insulated steam pipe in its thermally expanded position.
- (3) Conduits, junction boxes, pull boxes, termination cabinets, control stations, etc. shall have an 150 mm minimum clearances from the outside surface of any other piping than that mentioned above, ductwork, etc.

4.4.2 Conduit Bends (Elbows)

All “special” radius or field formed metallic conduit bends shall conform to the minimum radii shown in the tabulation below, and any deviations from the values shall be subject to review and approval by the Owner.

<u>Conduit Trade Size</u>		<u>Bend Radius</u>	
25	mm	305	mm
30	mm	460	mm
40	mm	460	mm
50	mm	610	mm

60 mm	610 mm
80 mm	915 mm
100 mm	915 mm
130 mm	1,250 mm
150 mm	1,250 mm

Bends for conduits containing two or more cables may be bent to the radius of the conduit size that would be used to carry the largest of the cables if that cable were in a conduit by itself.

4.4.3 Conduit Supports

Conduits that are run on walls, ceilings, beams, columns, etc., shall be securely supported with conduit supports spaced 2,700 mm maximum ,with at least one support between any two couplings. Conduit bends shall be supported on each side of the bend.

Metallic conduit systems between electrical equipment, such as between motors, junction boxes, cable trays, etc., shall be made up with tight connections to establish adequate electrical conductivity for ground fault current.

Conduits shall not be supported on the top surfaces of beam flanges. Conduits supported to the underside of beam flanges shall be located near the outer edges. Beam type conduit clamps shall not be permitted for conduit installations.

All field welds and all newly erected bare steel supports shall be and cleaned according to good welding practice, and shall be painted with suitable prime coat paint.

4.4.4 Conduit Terminations

Metallic conduits or connectors terminating at indoor sheet enclosures or equipment (except as specified herein otherwise), shall be provided with locknuts both inside and outside the enclosure, with insulating bushings on the ends of the conduits or connectors. Threadless, concrete tight, connectors and couplings may be used if made up tight, for electrical continuity.

Conduits shall be terminated at cable trays in accordance with the details of NFPA 70.

Metallic conduit terminations in outdoor or wet locations shall be made with suitable watertight fittings.

4.4.5 Conduit Pull Boxes and Expansion Fittings

Pull box fittings for EMT (electrical metallic tubing) shall be of the sleeve type. Pull box fittings exposed to the weather or expose splashing water shall be of the weatherproof sleeve type.

Pull boxes shall be sized on the following basis:

- (1) For straight pulls, the length of the box shall not be less than 24 times the outside diameter of the largest cable entering the box for unshielded cable and 48 times the outside diameter of the largest cable for shielded cable.
- (2) For angle or U pulls, the distance between a cable entering the box and its exit from the box shall be governed by the minimum bending radius of the cable.

The minimum bending radii to be used is 6 times the outside diameter for unshielded cable and 12 times the outside diameter for shielded cable. For angle pulls the minimum box sizes will be such that the opening is 18 times outside diameter for unshielded cable and 36 times the outside diameter for shielded cable on one (1) side.

- (3) Boxes shall conform to ANSI Standard C33.65 (UL-50) or equivalent. Galvanized coatings for the steel boxes shall conform to ASTM-A653.

Rigid type conduits connected to equipment mounted on the boiler or its supporting structure, subject to movement due to temperature variation, shall be provided with a length of liquidtight flexible metal conduit inserted in the rigid conduit run beyond the support where the conduit is fastened to the building wall or building steel.

4.4.6 Motor Terminal Housing Connections

For specific installation of conduit terminations to motors, equipment and devices, etc, method shall be provided by the Contractor for approval.

4.4.7 Conduit Bodies and Enclosures

Conduit bodies such as “LB” and “T,” shall not be used in conduits containing power or control cables except in special cases at conduit terminations where it is possible to strip off the jacket from the cable and make the necessary cable terminations without excessive bending that would cause damage to the cable insulation.

Gaskets shall be provided under the covers when conduit bodies and enclosures are installed in exposed locations, where water may enter and accumulate inside the conduit bodies or enclosures.

Junction box covers shall be fastened with zinc plated, slotted, hex head screws, and shall be gasketed.

4.4.8 Conduit in Floors and Walls

All openings in floors and walls that are left open for conduit installation shall be filled in with a sand, cement concrete mix by the Contractor. All conduits that are embedded in floors, walls, foundations, etc., shall be hot dipped galvanized rigid steel.

5. PERFORMANCE TESTING AND GUARANTEES

The conduits shall be subjected to the type and routine tests required by the applicable standard. The type tests for conduits shall include tests for ductility (bending) of the conduit and its zinc coating.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for conduits and fittings shall conform to the requirements included in Part 4, Subsection 2.7. The documentation as Specified shall include, but not be limited to, the followings:

- (1) Procedures for determining and thickness of galvanizing applied to conduits and fittings.
- (2) Electrical continuity test reports
- (3) All documentation required by applicable codes and standards.

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